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30 June 1976

Superseding
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2 June 1966

MILITARY SPECIFICATION

ANTENNA GROUP OE-4 ( )/GR

This specification is approved for use by U. S. Army Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the following units of one type of antenna group designated Antenna Group OE-4( )/GR:

Antenna Subassemblies (3.2) (3.5)

AS-2360( )/GR (Loop)
AS-2361( )/GR (Loop)
AS-2362( )/GR(Loop)
AS-2363( )/GR (Loop)
AS-2364( )/GR (Loop)
AS-2365( )/GR (H Adcock)
AS-2366( )/GR (Log Periodic)

Antenna Drives (3.2) (3.6)

TG-177( )/GR (Portable Pedestal)
TG-176( )/GR (Fixed Pedestal)

Mast Section Extension Shafts (3.2) (3.6)

AB-1113( )/GR
AB-1114( )/GR

Control, Antenna C-6183( )/GR (3.2) (3.7)

Indicator, Azimuth IP-740( )/GR (3.2) (3.8)

FSC 5895

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Battery Box Assembly CY-3589( )/GR (3.2) (3.9)

Cable Assemblies (3.2)

Vehicular Mounts (3.2)

Canvas Carrying and Stowage Cases (Field Packs) (3.2)

Compass Assembly (3.2)

1.2 Description. Antenna Group OE-4( )/GR consists of portable and shelter mountable directional antennas and ancillary units covering the frequency range of 0.535 to 1050 MHz. The antenna group is primarily used for visual and aural direction finding.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock(Container Grade), and Cut Shapes
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials and Processes; Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

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FEDERAL

FED-STD-356	Commercial Packing of Supplies and Equipment
FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461 w/Notice 4(EL)	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 w/Notice 3(EL)	Electromagnetic Interference Characteristics
MIL-STD-781	Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1285	Marking of Electrical and Electronic Parts

DRAWINGS

U.S. Army Electronics Command

DL-SM-A-643990	Antenna Group OE-4( )/GR
SC-A-46439	List of Accessories for Package Tester
SM-C-112622	Compass Magnetic, Surveyors (Modified)

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and the drawings listed in the following USAECOM Data Lists (including the requirements for parts, materials and processes thereon).

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DATA LIST	EQUIPMENT
DL-SM-A-643990	Antenna Group OE-4( )/GR
DL-SM-A-642960	Leg, Electrical Equipment MX-6334( )/GR
DL-SM-A-642984	Drive, Antenna TG-177( )/GR
DL-SM-A-643040	Drive, Antenna TG-176( )/GR
DL-SM-A-643070	Antenna AS-2365( )/GR
DL-SM-A-643160	Mast Section AB-1113( )/GR
DL-SM-A-643170	Mast Section AB-1114( )/GR
DL-SM-A-643190	Antenna AS-2366( )/GR
DL-SM-A-643290	Antenna Control C-6183( )/GR
DL-SM-A-643490	Antenna AS-2360( )/GR
DL-SM-A-643500	Antenna AS-2361( )/GR
DL-SM-A-643510	Antenna AS-2362( )/GR
DL-SM-A-643520	Antenna AS-2363( )/GR
DL-SM-A-643530	Antenna AS-2364( )/GR
DL-SM-A-643600	Indicator, Azimuth IP-740( )/GR
DL-SM-A-643630	Base, Shock Mount MT-4132( )/GR
DL-SM-A-643640	Base, Shock Mount MT-4133( )/GR
DL-SM-A-644410	Battery Box Assy CY-3589( )/GR
DL-SM-A-690149	Compass Assy
DL-SM-A-690150	Field Pack, Canvas CW-1056( )/GR
DL-SM-A-690151	Field Pack, Canvas CW-1057( )/GR
DL-SM-A-690152	Field Pack, Canvas CW-1058( )/GR
DL-SM-A-690153	Field Pack, Canvas CW-1059( )/GR
DL-SM-A-690154	Field Pack, Canvas CW-1060( )/GR
DL-SM-A-690155	Field Pack, Canvas CW-1061( )/GR
DL-SM-A-690156	Field Pack, Canvas CW-1062( )/GR
DL-SM-A-690157	Field Pack, Canvas CW-1063( )/GR
DL-SM-A-690158	Field Pack, Canvas CW-1064( )/GR
DL-SM-A-690159	Cable Assy CX11950( )/U
DL-SM-A-690160	Cable Assy CX11951( )/U
DL-SM-A-690161	Cable Assy CX11952( )/U
DL-SM-A-690162	Cable Assy CX11953( )/U
SC-DL-157329	Cable Assy, RF, CG-409G( )/U

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3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.3.1 Wiring. All wiring shall be teflon coated.

3.3.2 Fabric and thread. Fabric and thread shall be preshrunk or allowance shall be made for shrinkage in order to provide for satisfactory fit of finished items both before and after they are soaked in tap water at normal room temperature for 24 hours and then dried thoroughly at normal room temperature (See 4.19).

3.3.3 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings.

3.4 Marking.

3.4.1 General. Marking shall conform to MIL-M-13231 and MIL-STD-1285. (see 4.4). Front panel marking shall be Group I as described in MIL-M-13231.

3.4.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.3 Serial numbers. The following items shall have serial numbers:

Antenna Subassembly AS-2360( )/GR (Loop)
Antenna Subassembly AS-2361( )/GR (Loop)
Antenna Subassembly AS-2362( )/GR(Loop)
Antenna Subassembly AS-2363( )/GR(Loop)
Antenna Subassembly AS-2364( )/GR(Loop)
Antenna Subassembly AS-2365( )/GR (H Adcock)
Antenna Subassembly AS-2366( )/GR (Log Periodic)
Antenna Drive TG-176( )/GR (Fixed Pedestal)
Antenna Drive TG-177( )/GR (Portable Pedestal)
Antenna Control C-6183( )/GR
Azimuth Indicator IP-740( )/GR

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TABLE I - ANTENNA SUBASSEMBLIES

Nomenclature	Antenna Type	Band No. (Note 1)	Frequency Range (MHz)	Direct to-reciprocal Bearing Error (Degrees)	Minimum Sensitivity (field strength uv/m)	Receiver Nomenclature
AS-2360()/GR	Shield loop with monopole whip sense	1	0.535 to 1.15	2	5	R-1218
		2	1.10 to 2.35	2	5	
		3	2.25 to 4.80	2	2	
		4	4.60 to 9.90	2	2	
		5	9.50 to 20.5	2	20	
AS-2361()/GR	Shielded loop with monopole whip sense	5	9.50 to 20.5	2	2	R-1218
AS-2362()/GR	Shielded loop with monopole whip sense	6	19.5 to 33.0	2	3	R-1518
AS-2363()/GR	Shielded loop with monopole whip sense	7	31 to 56	2	5	R-1518
AS-2364()/GR	Shielded loop with monopole whip sense	8	55 to 101	2	5	R-1518
AS-2365()/GR	H Adcock array	-	95 to 210	2	-	R-1518 and/ or R-902
AS-2366()/GR	Log periodic array	-	190 to 1050	-	-	R-902

Note 1 - Indicates band numbers of Antenna Control C-6183()/GR.

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3.5 Antenna subassemblies. The seven directional antenna subassemblies listed in Table I shall exhibit directional properties that allow azimuthal determination of unknown signal sources.

3.5.1 Frequency range. The antenna subassemblies shall operate over the frequency ranges specified in Table I. Antenna subassembly AS-2360( )/GR shall cover its frequency range in 5 bands as listed in Table I. The remaining six antenna subassemblies shall each cover their frequency range in single bands (see 4.7.14).

3.5.2 Antenna pattern. All antenna subassemblies, except the log periodic AS-2366( )/GR, shall exhibit a symmetrical figure "8" type response pattern, with null depth of not less than 25db (see 4.7.2).

3.5.3 Direct-to-reciprocal bearing error. All antenna subassemblies, except the log periodic, shall exhibit a direct-to-reciprocal bearing error not exceeding the values specified in Table I (see 4.7.6).

3.5.4 Bearing accuracy. The bearing error obtained by all antenna subassemblies, except the log periodic, shall be less than $\pm$ 2 degrees over their respective frequency ranges (see 4.7.15).

3.5.5 Antenna Subassemblies (AS-2360( )/GR through AS-2364( )/GR. The loop antenna subassemblies shall receive vertically polarized signals (see 4.7.2).

3.5.5.1 Sensitivity. The loop antenna subassemblies, when oriented for maximum signal response, shall not require more than the signal field strength listed in Table I to produce a 6db signal-plus-noise to noise receiver output (see 4.7.16).

3.5.5.2 Band switching (AS-2360( )/GR). Using Antenna Control C-6183( )/GR, the elapsed time between initiating band change and actuation of the band switch relay shall not exceed 3 seconds (see 4.7.22).

3.5.5.3 Frequency tuning. When supplied with control voltages from Control C-6183( )/GR, loop antenna subassemblies AS-2360( )/GR through AS-2364( )/GR shall resonate over their specified frequency ranges (see 4.7.5).

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3.5.5.4 Sense element. The sense element is a vertical monopole whip. The sense mode, controlled by the antenna control unit through the sense relay of the antenna subassembly, shall provide a null shift of not less than 15 degrees from the unshifted axis, without reversal, throughout the specified operating frequency range (see 4.7.7).

3.5.5.5 Output impedance. The output impedance of the RF amplifiers shall be a 50 ohms ± 10% (see 4.7.12).

3.5.5.6 Dynamic range. The RF amplifier shall have a dynamic range that enables use of the equipment with any received signal from the minimum sensitivity level (Table 1) to 0.25 volt/meter. This requirement may be accomplished with the use of the gain control on the antenna control unit (see 4.7.18).

3.5.5.7 Intermodulation distortion products. The intermodulation distortion products shall be suppressed a minimum of 70 db (see 4.7.19).

3.5.5.8 Envelope distortion. There shall be a maximum of one percent difference in the envelope distortion between the loop antennas and a dipole antenna when tested as specified in 4.7.20.

3.5.6 Antenna Subassembly AS-2365( )/GR (H Adcock).

3.5.6.1 Output impedance. The output impedance shall be 50 ohms ± 10% (see 4.7.12).

3.5.6.2 Gain. The antenna subassembly shall have a gain equal to or exceeding that of a tuned dipole (see 4.7.17).

3.5.6.3 Sense. The sense mode shall provide a direct-to-reciprocal null displacement of not less than 15 degrees, without reversal, throughout the specified operating frequency range (see 4.7.7).

3.5.6.4 Polarization. The antenna subassembly shall receive polarized signals at full horizontal and full vertical aspect. Both electrical and mechanical stops shall limit the tilt of the antenna array through the required 90 ± 5 degrees horizontal to vertical position (see 4.7.21).

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3.5.7 Antenna AS-2366( )/GR (Log Periodic). The antenna subassembly is a log periodic array arranged for a 45 degree polarization. It shall utilize an infinite type balun with an average VSWR of not greater than 3.5:1 and a maximum VSWR not greater than 4:1 at any frequency when measured at the output of the rotary joint of the antenna subassembly (see 4.7.13). The output impedance shall be 50 ohms ± 10% (4.7.12). This array shall receive polarized signals from the horizontal through the vertical aspect (see 4.7.3).

3.5.7.1 Beam pattern. The main lobe of the antenna pattern shall have an average beamwidth of not greater than 90 degrees and a maximum beamwidth not greater than 115 degrees at any frequency, when measured as described in 4.7.3. The 6 db down points shall be used to determine beamwidth. The beam tilt, when measured as described in 4.7.3, shall average not greater than 3 degrees and the maximum beam tilt shall be not greater than 5 degrees at any frequency, when receiving either horizontally or vertically polarized signals. Side and rear lobes shall be suppressed at least an average of 10 db and not less than 6 db below the main lobe at any one frequency (see 4.7.3).

3.5.7.2 Gain. The antenna shallhave an average gain of at least 3 db over a tuned dipole and a gain of not less than 2.5 db at any frequency, when measured as described in 4.7.17, with the antenna in its normal operating position and with the target signals in either full horizontal or full vertical polarization.

3.6 Antenna drives and extension shafts.

3.6.1 Antenna Drive TG-177( )/GR (Portable Pedestal). This unit shall rotate the antenna and shaft at a rate of 60 ± 5 rpm during visual operation. The unit shall operate at 0-6 ± 0.5 rpm either clockwise or counterclockwise in the remote aural mode (see 4.7.11).

3.6.2 Antenna Drive TG-176( )/GR (Fixed Pedestal). This unit shall rotate the antenna and shaft at a rate of 60 ± 5 rpm during visual mode of operation (see 4.7.11).

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3.6.3 Mast Section Extension Shafts AB-1113( )/GR and AB-1114( )/GR. The extension shafts shall carry signal and control functions between the antenna drives and antenna subassemblies. The extension shafts shall not introduce a bearing shift error in excess of 1/4 degree from the bearing determined when the antenna subassembly is mounted directly on the antenna drive (see 4.7.2.6).

3.7 Antenna Control C-6183( )/GR. This unit shall control antenna drive rotation, frequency tuning, band switching, sense, polarization and amplifier gain, and shall meet the requirements of 3.7.1 through 3.7.7.

3.7.1 Power converter. An internal DC to DC regulated power converter shall convert a 24 $\pm$ 0.5 VDC input to the voltages required for the antenna amplifiers and frequency tuning as follows: (see 4.7.10.1).

(a) Filament supply (regulated) (1.50 $\pm$ 0.02 volts).

(b) Plate supply (36 $\pm$ 2 volts) and loop gain supply voltage (0-36 $\pm$ 2 volts).

(c) Bandswitch supply voltage (300 $\pm$ 5.0 volts).

(d) Frequency tuning supply voltage (regulated) (0-101 $\pm$ 2 volts).

3.7.2 DC power. Control C-6183( )/GR shall operate in the portable, local and aural modes of operation (see 4.7.11).

3.7.3 Azimuth motor control switch. The switch shall provide selection of standard speed (60 $\pm$ 5 rpm) continuous unidirectional rotation of the antenna drive for visual mode operation at the 24 volts $\pm$ 0.5 DC input, or controlled low-speed rotation for aural (manual search) mode operation (see 4.7.11).

3.7.4 Azimuth motor manual control. The forward-reverse switch and speed control shall regulate Antenna Drive TG-177( )/GR direction and speed (0-6 $\pm$ 0.5 rpm) during aural mode operation. The forward-reverse switch shall be activated only when the azimuth motor control switch is in the controlled low-speed position (see 4.7.11).

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3.7.5 Antenna frequency tuning control. The tuning control shall regulate voltage applied to the voltage-tuned diode capacitors of loop antenna subassemblies AS-2360( )/GR through AS-2364( )/GR. The frequency tuning dial shall be accurate to ± 5% of the actual antenna resonant frequency (see 4.7.5).

3.7.6 Sense switch. The sense switch shall provide operation of the sense relays in the AS-2360( )/GR thru AS-2364( )/GR and H Adcock AS-2365( )/GR antenna subassemblies for a period of 1 to 4 seconds to eliminate the 180 degrees ambiguity from the antenna pattern (see 4.7.7).

3.7.7 Polarization tilt. The momentary on-off toggle switch shall provide manual control of the rotary actuator of the AS-2365( )/GR H Adcock antenna subassembly (see 4.7.21).

3.8 Azimuth Indicator IP-740( )/GR.

3.8.1 Display. When the log periodic display switch is in the "OFF" position, the no-signal display shall be a filled-in circle derived from the antenna drive resolver. There shall be no observable circumferential flattening or other distortion on the circle. The bearing pattern for the null type antennas shall be propeller-shaped with the tips corresponding to the direct and reciprocal antenna nulls. The tips shall be sharp and distinct with no blurring. Sense pattern shall be a bent-propeller shape, with the tips bent away from the direct bearing. When the log periodic display switch is in the "ON" position, the no-signal shall be a dot. The log periodic display shall be a tear shaped lobe corresponding to the maximum response of the antenna (see 4.7.8).

3.8.2 Visual bearing deviation. The null bearing on the azimuth indicator with the antenna rotating at 60 rpm shall not deviate more than 1.5 degrees from the null bearing on the azimuth indicator at 0 rpm for Antenna Subassemblies AS-2360( )/GR through AS-2365( )/GR, inclusive (see 4.7.9).

3.8.3 Power converter. The internal DC to DC power converter shall convert the 24 volts DC input to 1.5 ± 10% volts DC for the CRT filament and 3000 ± 250 volts DC for the CRT high voltage (see 4.7.10.2).

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3.8.4 Log periodic display switch. The on-off switch shall allow indicator operation with maximum type as well as null type antennas. This switch in the "ON" position shall provide blanking and cause the pattern size to go to zero deflection with no diode input voltage (see 4.7.8.3).

3.8.5 Blanking circuitry. Blanking circuitry shall enable half of the sweep to be blanked so that only a line from the center of the CRT to one side is usable when the sense switch of the antenna control unit is actuated, with the log periodic display switch in the "OFF" position (see 4.7.7).

3.8.6 Shape control. The shape control shall provide adjustment of the amount of horizontal size available to the CRT. Adjustment of this control shall permit the achievement of a circular display on the CRT when no receiver diode voltage is being fed to the indicator (see 4.7.8).

3.9 Battery Box Assembly CY-3589( )/G. The box assembly shall provide a watertight inclosure for Batteries BA-411/U and shall be constructed in accordance with DL-SM-A-644410.

3.10 Input voltage. The equipment shall be capable of operation from a 24 VDC input voltage, which may vary from 20 to 32 volts DC with a 0.1 volt ripple. The circuits shall not be damaged by accidental application of reverse polarity (see 4.7.4).

3.11 Service conditions. The equipment shall meet the following service conditions, where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.11.1 Temperature.

3.11.1.1 Low. During testing as specified in 4.8.1, the equipment, less the canvas cases, shall meet the performance specified in Table VII. After testing, the equipment shall meet the performance specified in Tables IV and V.

3.11.1.2 High. During testing as specified in 4.8.2, the equipment, less the canvas cases, shall meet the performance specified in Table VII. After testing, the equipment shall meet the performance specified in Tables IV and V.

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3.11.2 Humidity. The equipment, including the canvas cases, shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the performance specified in Table VII during the operating times specified in 4.9. After testing, the equipment shall meet the performance specified in Tables IV and V.

3.11.3 Altitude (elevation). During testing, as specified in 4.10, the equipment, less the canvas bags, shall meet the performance specified in Table VII. After testing, the equipment shall meet the performance specified in Tables IV and V.

3.11.4 Leakage (immersion). Battery Box CY-3589( )/GR shall show no evidence of leakage after tested as specified in 4.17.

3.11.5 Orientation. When tested as specified in 4.18, the equipment shall meet the performance specified in Table VII.

3.11.6 Rain. After testing as specified in 4.11, the equipment, less the canvas cases, shall meet the performance specified in Table VII.

3.11.7 Dust. After testing as specified in 4.12, the equipment, less the canvas cases, shall meet the performance specified in Tables IV and V.

3.11.8 Vibration, internal. With the shock mounts (if any) blocked, the amplitude of vibration of any part, subassembly or structural member of Antenna Drive TG-176( )/GR, Azimuth Indicator IP-740( )/GR and Antenna Control C-6183( )/GR shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hertz (see 4.13).

3.11.9 Bounce and shock. The equipment shall meet the requirements of Table II.

3.11.10 Salt fog. After the salt fog test of 4.21, the equipment, including the canvas cases, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface; or exfoliation (see 6.4) of metal. Electrical operation is not required.

3.11.1 Fungus. After the fungus test of 4.22, the equipment, including the canvas cases shall be visually examined, using a 10-power magnifier.

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TABLE II - BOUNCE AND SHOCK

TEST	TEST PARAGRAPH	PERFORMANCE AFTER TEST
Bounce; Class L units, vehicular mounted: Antenna Control C-6183( )/GR Azimuth Indicator IP-740( )/GR Antenna Drive TG-176( )/GR	4.14.1	Specified performance (Note A). No physical damage.
Bounce; Class F units field transportable: Antenna Subassemblies AS-2360( )/GR thru AS-2366( )/GR Antenna Drive (portable) TG-177( )/GR Mast Section Extension Shafts AB-1113( )/GR and AB-1114( )/GR Antenna Control C-6183( )/GR Azimuth Indicator IP-740( )/GR Battery Box Assembly CY-3589( )/GR	4.14.2	Specified performance (Note A). No physical damage except minor surface abrasion.
Shock; bench handling: Antenna Control C-6183( )/GR Azimuth Indicator IP-740( )/GR	4.15	Specified performance (Note A). No physical damage.
Shock; drop: All equipment, excluding Antenna Drive TG-176( )/GR and the Vehicular Mounts, in their canvas carrying and stowage cases.	4.16	Operable (Note A). Any physical damage shall be minor only.

Note A. The equipment shall meet the performance specified in Tables IV and V

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The equipment shall show no more than six minute unrelated spots each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.4) as evidenced by growth colonization (see 6.4) (which includes branching (see 6.4) and sporulation (see 6.4) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.4) shall not be included in this evaluation.

3.12 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment and shall conform to requirement 7 of MIL-STD-454 (see 4.23). Individual items shall not be hand-picked for fit or performance, however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating or characteristic.

3.13 Preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (also see 4.5).

3.14 Burn-In. Each unit on order shall meet the requirements herein, without subsequent processing, after subjection to burn-in as specified in 4.6.2.

3.15 Electromagnetic interference (EMI) characteristics. The equipment shall meet the EMI requirements specified in MIL-STD-461, Notice 4(EL), Table A-I-MIL-E-55301, Class CE (see 4.20).

3.16 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In MIL-P-11268

General requirements for plastic material and plastic parts
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Process for soldering
Cleaning
Riveting
General requirements for securing of parts
Cleaning after welding
Mechanical operation of controls
Process for welding
Gears

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Flux and cleaning agent for soldering	Self-tapping screws

(b) In this specification:

3.3.1 Wiring
3.3.3 Finish

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First Article inspection (see 4.3) does not include preparation for delivery.

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.25).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 First article units. The contractor shall furnish four (4) first articles of the complete Antenna Group OE-4( )/GR.

4.3.2 First article inspection. The first article inspection shall consist of the inspection specified in Table III and shall be performed in the order specified therein. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspection.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

Item	Where required
Parts, Materials and Processes - general	3.3
Finish	3.3.3
Marking	3.4

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4, preconditioning 4.6.1, burn-in 4.6.2 and 4.5.1 through 4.5.4 in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table IV. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table IV.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

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TABLE III - First Article Inspection

Inspections	Rqmt Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4
1. Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units.			
2. Bounce preconditioning	3.13	4.6.1	Inspections to be performed on all units			
3. Burn-in preconditioning	3.14	4.6.2	Inspections to be performed on all units			
4. Group A inspection	See Table IV		Inspections to be performed on all units			
5. Group B inspection	See Table V		Inspections to be performed on all units			
6. Group C inspection (electrical)	See Table VI		All electrical inspections listed in Subgroup I of Table VI are to be performed on all units			
7. EMI	3.15	4.20	1			
8. Environmental						
Temperature	3.11.1	4.8		1		
Humidity	3.11.2	4.9		2		
Altitude	3.11.3	4.10			1	
Immersion	3.11.4	4.17			3	
Orientation	3.11.5	4.18			2	
Rain	3.11.6	4.11			4	
Dust	3.11.7	4.12				2
Vibration	3.11.8	4.13				1
Bounce, Class L units	3.11.9	4.14.1	2			
Bounce, Class F units	3.11.9	4.14.2		3		
Shock, bench handling	3.11.9	4.15	3			
Shock, drop	3.11.9	4.16	4			
Salt, fog	3.11.10	4.21		4		
Fungus	3.11.11	4.22				3

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a. Inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. Numbers in the unit columns opposite inspections 7 and 8 specify the order of inspection for the indicated unit.

TABLE IV - GROUP A INSPECTION

Inspections	Requirement Para.	Test Para.	AQL Major	AQL Minor
Visual and Mechanical				
Antenna Subassemblies				
AS-2360( )/GR (loop)	3.16	4.24	1.0%	4.0%
AS-2361( )/GR (loop)	3.16	4.24	1.0%	4.0%
AS-2362( )/GR (loop)	3.16	4.24	1.0%	4.0%
AS-2363( )/GR (loop)	3.16	4.24	1.0%	4.0%
AS-2364( )/GR (loop)	3.16	4.24	1.0%	4.0%
AS-2365( )/GR (H Adcock)	3.16	4.24	1.0%	4.0%
AS-2366( )/GR (Log Periodic)	3.16	4.24	1.0%	4.0%
Antenna Drive				
TG-176( )/GR (fixed pedestal)	3.16	4.24	1.0%	4.0%
TG-177( )/GR(portable pedestal)	3.16	4.24	1.0%	4.0%
Mast section extension shafts				
Extension Shaft AB-1113( )/GR (metal)	3.16	4.24	1.0%	4.0%
Extension Shaft AB-1114( )/GR (dielectric)	3.16	4.22	1.0%	4.0%
Antenna Control C-6183( )/GR	3.16	4.24	1.0%	4.0%
Azimuth Indicator IP-740( )/GR	3.16	4.24	1.0%	6.5%

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Battery Box CY-3589( )/GR	3.16	4.24	1.0%	4.0%
Cable assemblies	3.16	4.24	1.0%	4.0%
Bases, shock mount	3.16	4.24	1.0%	4.0%
Canvas field packs	3.16	4.24	1.0%	4.0%
Electrical				
Antenna Subassemblies				
Direct-to-reciprocal bearing error	3.5.3	4.7.6, 4.7.2.4	1.0%	*
Frequency tuning	3.5.5.3	4.7.5	1.0%	*
Sense element (loops)	3.5.5.4	4.7.7	1.0%	*
Sense (H Adcock)	3.5.6.3	4.7.7	1.0%	*
Polarization (H Adcock)	3.5.6.4	4.7.21	1.0%	*
Antenna Drives				
Antenna Drive TG-176( )/GR drive motor (fixed pedestal)	3.6	4.7.11	1.0%	*
Antenna Drive TG-177( )/GR drive motor (portable pedestal)	3.6	4.7.11	1.0%	*
Antenna Control C-6183( )GR				
Power converter	3.7.1	4.7.10.1	1.0%	*
Azimuth motor control switch	3.7.3	4.7.11	1.0%	*
Azimuth motor manual control	3.7.4	4.7.11	1.0%	*
Antenna frequency tuning control	3.7.5	4.7.5	1.0%	*
Frequency tuning dial calibration	3.7.5	4.7.5	1.0%	*
Sense switch	3.7.6	4.7.7	1.0%	*
Polarization tilt	3.7.7	4.7.21	1.0%	*
Azimuth Indicator IP-740( )/GR				
Display	3.8.1	4.7.8	1.0%	*
Visual bearing deviation	3.8.2	4.7.9	1.0%	*
Power converter	3.8.3	4.7.10.2	1.0%	*
Blanking circuitry	3.8.5	4.7.7	1.0%	*
Log periodic display switch	3.8.4	4.7.8.3	1.0%	*

* All electrical defects shall be considered major.

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4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table V and to the special levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plan. The Group B sampling plans, for the AQL's listed in Table V, shall be as follows:

AQL	Inspection level
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within Group B. The inspection shall be performed in any order which is satisfactory to the Government.

TABLE V - GROUP B INSPECTION

Inspection	Requirement Para.	Test Para.	AQL (see 4.5.2.1)
Dimensional interchangeability	3.12	4.23	6.5%
Envelope distortion (Loops)	3.5.5.8	4.7.20	4.0%
VSWR (Log Periodic)	3.5.7	4.7.13	4.0%
Output impedance (Loops)	3.5.5.5	4.7.12	4.0%
Output impedance (H Adcock)	3.5.6.1	4.7.12	4.0%
Output impedance (Log Periodic)	3.5.7	4.7.12	4.0%
Band switch	3.5.5.2	4.7.22	4.0%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1. The inspection shall consist of the inspections specified in Table VI.

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TABLE VI - GROUP C INSPECTION

Inspection	Requirement Para.	Test Para.
Subgroup I		
Target Polarization	3.5.5, 3.5.6.4	4.7.2.5
Mast Sections	3.6.3	4.7.2.6
Beam Pattern	3.5.7, 3.5.7.1	4.7.3
Frequency Range (Loops)	3.5.1	4.7.14
Bearing Accurracy (Loops & H Adcock)	3.5.4	4.7.15
Sensitivity (Loops)	3.5.5.1	4.7.16
Gain (H Adcock)	3.5.6.2	4.7.17
Gain (Log Periodic)	3.5.7.2	4.7.17
Dynamic Range (Loops)	3.5.3, 3.5.4, 3.5.5.4, 3.5.5.6	4.7.18
Intermodulation Distortion Products	3.5.5.7	4.7.19
Accidental Reverse Polarity	3.10	4.7.4
Subgroup II		
Vibration		
Resonance Search	3.11.8	4.13
Bounce, Class L and Class F	3.11.9	4.14
Altitude	3.11.3	4.10
Salt Fog*	3.11.10	4.21
Subgroup III		
Temperature	3.11.1	4.8
Humidity	3.11.2	4.9
Immersion	3.11.4	4.17
Subgroup IV		
Shrinkage	3.3.2.1	4.19
EMI	3.15	4.20
Fungus*	3.11.11	4.22

* The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

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4.5.3.1 Sampling for Group C inspection. Four samples of the complete equipment shall be selected at random, 1 sample for each (Subgroup I through and including Subgroup IV) in Table VI. The first samples shall be selected at the start of production from the first quality conformance inspection lot. These samples shall constitute the Group C requirement for the first 50 units produced. Thereafter, 4 samples of the complete equipment shall be selected at random, 1 sample for each subgroup I through Subgroup IV in Table VI from every 100 units and fraction thereof produced. Salt fog and fungus shall not be repeated after the first production lot Group C inspection.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in an order satisfactory to the Government.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided that they are resubjected to and pass Group A inspection after repair of all damage.

4.6 Bounce Preconditioning/Burn-in.

4.6.1 Bounce preconditioning. All equipment, with the exception of Antenna Control C-6183( )/GR and Azimuth Indicator IP-740( )/GR, shall be stowed in their respective canvas carrying and stowage cases, and then placed on the table of the Package Tester, Type 1000-SC, as made by L. A. B. Corporation, Skaneateles, New York, or equal. The package tester shall be adjusted to a speed of 284 $\pm$ 2 rpm. The equipment shall be subjected to this preconditioning for 1 minute. The Antenna Control and Azimuth Indicator shall be placed in their normal operating position on the package tester. The package tester, shafts in phase, shall have a speed such that it is possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment to distance of 3 inches as the equipment is being bounced. The equipment shall be subjected to this preconditioning for 1 minute. After bounce preconditioning none of the equipment shall be repaired, aligned, cleaned or otherwise changed prior to subjection to acceptance inspection.

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4.6.2 Burn-in. All electronic equipment of each unit shall be subjected to a 48 hour burn-in period. The last 12 hours of burn-in shall be failure free. The burn-in shall be conducted under room ambient conditions with an input voltage of 24 $\pm$ 0.5 VDC. All failures shall be recorded and analyzed. Each measurement shall be monitored for intermittencies for a period of one (1) minute. A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this specification or the contractor's approved test procedures. Paragraph 5.5 thru 5.10 inclusive of MIL-STD-781 shall apply. The types of failures and required actions shall be in accordance with paragraph 5.5.1 of MIL-STD-781. Determination of compliance shall be accordance with paragraph 5.4.8 of MIL-STD-781.

4.7 Electrical tests.

4.7.1 Standard test conditions. Unless otherwise specified, the electrical tests shall be conducted under the following conditions:

(a) Temperature and Humidity ---------- room ambient

(b) Input voltage ---------- 24 $\pm$ 0.5 VDC

(c) Receiver rf input impedance ---------- 50 ohms $\pm$ 1%

4.7.2 Antenna pattern measurements (loops and H Adcock).

4.7.2.1 Antenna test site. Antenna pattern measurements shall be performed in an open field area so located, as to be as free as possible of man-made noise. The field shall be a flat and level cleared area exceeding 250 feet x 500 feet. The target antenna and the receiving antenna shall have a maximum separation of 300 feet for the MF and HF frequency ranges. The separation of all other frequencies shall be greater than 2 wave lengths. There shall be no electrical lines between the target antenna and the receiving antenna (see Fig. 1 of SM-A-690323, DL-SM-A-643990).

4.7.2.2 Target signal generator. A calibrated signal generator capable of being continuously varied in output, together with a suitable antenna, shall be used as a radiating target. The target generator shall operate from an isolated power source located at the transmitting site.

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4.7.2.3 Field strength meter. The receiving field strength meter shall be a calibrated Empire Noise and Field Intensity Meter, Model NF-105, or equal, having an input impedance of 50 ohms.

4.7.2.4 Null depth. Measure the depth of the two nulls of each antenna. The null depth shall meet the requirements of 3.5.2. The direct-to-reciprocal bearing shall meet the requirements of 3.5.3 when tested in accordance with paragraph 4.2 of SM-A-690323. The null shift shall conform to 3.5.5.4 or 3.5.6.3 when tested in accordance with paragraph 4.2 of SM-A-690323.

4.7.2.5 Target polarization. Antenna Subassemblies AS-2360( )/GR through AS-2364( )/GR inclusive, shall be tested using vertically polarized target signals. Antenna Subassembly AS-2365( )/GR shall be tested using vertically polarized and horizontally polarized target signals. The antenna subassemblies shall meet the polarization requirements of 3.5.5 or 3.5.6.4 respectively when tested in accordance with paragraph 6.0 of SM-A-690323

4.7.2.6 Mast sections. Antenna Subassembly AS-2361( )/GR shall be tested with and without two Extension Shafts AB-1114( )/GR. Antenna Subassembly AS-2363( )/GR shall be tested with and without two Extension Shafts AB-1113( )/GR. These tests shall meet the requirements of 3.6.3.

4.7.3 Beam pattern measurements (Log Periodic). The beam pattern measurements shall be performed in an open field area as specified in 4.7.2.1. Employing the field strength meter specified in 4.7.2.3, relative voltage beam patterns shall be obtained. The target power level shall be such that the field strength at the receiving antenna location is sufficient to cause the nulls of the beam pattern to correspond to the greater of either one microvolt or the signal required for a 10 db signal-plus-noise to noise ratio at the receiver output. Beam patterns shall be made over the frequency range of 190 to 1050 MHz in 50 MHz steps for Antenna Subassembly AS-2366( )/GR. Readings shall be obtained at least every 10 degrees. Beam pattern measurements shall meet the requirements of 3.5.7 and 3.5.7.1.

4.7.4 Input voltage test. Apply 20 $\pm$ 0.5 VDC to the equipment and perform the tests specified in Table VII. Repeat at 32 $\pm$ 0.5 VDC. Apply 24 $\pm$ 0.5 VDC with the polarity reversed. The requirements of 3.10 shall be met.

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TABLE VII - Abbreviated System Operation

Test	Requirement Para.	Test Para.
Tuning (Note A)	3.5.5.3, 3.7.5	4.7.5
Sense	3.5.5.3, 3.5.6.3	4.7.7
Azimuth indicator display	3.8.1	4.7.8
Visual bearing deviation	3.8.2	4.7.9
Voltage measurements	3.7.1, 3.8.3	4.7.10
Antenna drive rotation	3.6, 3.7.3, 3.7.4	4.7.11
H Adcock tilt	3.5.6.4	4.7.21
Band switch	3.5.5.2	4.7.22

Note A. Deterioration of frequency tuning dial accuracy due to temperature cycling shall not exceed ± 10% of the actual antenna resonant frequency.

4.7.5 Frequency tuning test (Loops). Feed the output of a calibrated signal generator into a whip antenna (or couple directly into the loop). The signal generator shall provide a signal with sufficient strength to overcome ambient noise and shall be capable of operating over the frequency range of 0.5 to 101 MHz. Antenna Subassemblies AS-2360( )/GR through AS-2364( )/GR shall be tuned to each test frequency specified in Table VIII using the tuning control of Antenna Control C-6183( )/GR and shall meet the tuning requirements of 3.5.5.3 and 3.7.5 when tested per paragraph 5.0 of SM-A-690323.

4.7.6 Direct-to-reciprocal bearing error test (Loops and H Adcock). Feed the output of a calibrated signal generator, as specified in 4.7.2.2, into a whip antenna. The direct-to-reciprocal bearing error of Antenna Subassemblies AS-2360( )/GR through AS-2365( )/GR shall be determined at the test frequencies specified in Table VIII using the display on Azimuth Indicator IP-740( )/GR. The direct-to-reciprocal bearing error shall meet the requirements of 3.5.3. The loops shall be tested using vertically polarized target signals. The H Adcock shall be tested using vertically and horizontally polarized target signals in accordance with paragraph 4.2 of SM-A-690323.

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TABLE VIII - TEST FREQUENCIES

Antenna Subassembly	Band No. (Note 1)	Test Frequencies (MHz)	Receiver
AS-2360( )/GR (Loop)	1	0.535, 0.85, 1.15	R-1218
AS-2360( )/GR (Loop)	2	1.10, 1.75, 2.35	R-1218
AS-2360( )/GR (Loop)	3	2.25, 3.50, 4.80	R-1218
AS-2360( )/GR (Loop)	4	4.60, 7.25, 9.90	R-1218
AS-2360( )/GR (Loop)	5	9.50, 15.0, 20.5	R-1218
AS-2361( )/GR (Loop)	5	9.50, 15.0, 20.5	R-1218
AS-2362( )/GR (Loop)	6	19.5, 26.0, 33.0	R-1518
AS-2363( )/GR (Loop)	7	31, 44, 56	R-1518
AS-2364( )/GR (Loop)	8	55, 78, 101	R-1518
AS-2365( )/GR (H Adcock)	-	100, 150, 200	R-1518 R-902
AS-2366( )/GR (Log Periodic)	-	200, 300, 400	R-902
		600, 800, 1000	(Note 2)

Note 1. Corresponding to band numbers of Antenna Control C-6183( )/GR.

Note 2. The contractor shall supply a suitable receiver for test purposes capable of operating over the frequency range of 400 to 1000 MHz.

4.7.7 Sense test (Loops and H Adcock). Feed the output of a calibrated signal generator, as specified in 4.7.2.2, into a whip antenna. The sense of Antenna Subassemblies AS-2360( )/GR through AS-2365( )/GR, inclusive, shall be determined at the test frequencies specified in Table VII using the display on Azimuth Indicator IP-740( )/GR. The sense shall meet the requirements of 3.5.5.4, 3.5.6.3, 3.7.6 and 3.8.5 for the respective antennas when tested in accordance with paragraph 10.0 of SM-A-690323. The loops shall be tested using vertically polarized target signals. The H Adcock shall be tested using vertically and horizontally polarized target signals.

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4.7.8 Azimuth Indicator display test.

4.7.8.1 The test set-up shall be in accordance with Figure 7 of SM-A-690323.

4.7.8.2 Null type display. The null type displays shall meet the requirements of 3.8.1 when tested in accordance with paragraph 10.0 of SM-A-690323.

4.7.8.3 Maximum type display. When testing Antenna Subassembly AS-2366( )/GR at 190, 300 and 400 MHz per paragraph 10.0 of SM-A-690323, the requirements of 3.8.1, 3.8.4 and 3.24 shall be met.

4.7.9 Visual bearing deviation test. The azimuth of the null bearing on Azimuth Indicator IP-740( )/GR with the antenna rotating at 60 rpm shall meet the requirements of 3.8.2. This test shall be performed on Antenna Subassemblies AS-2360( )/GR through AS-2365( )/GR inclusive, at the test frequencies specified in Table VIII per paragraph 4.2 of SM-A-690323.

4.7.10 Voltage measurements.

4.7.10.1 Power converter (Antenna Control). Measure the output voltages of the internal DC to DC regulated power converter of Antenna Control C-6183( )/GR per paragraph 8.0 of SM-A-690323. These measurements shall meet the requirements of 3.7.1.

4.7.10.2 Power converter (Azimuth Indicator). Measure the output voltages of the internal DC to DC power converter of Azimuth Indicator IP-740( )/GR per paragraph 9.0 of SM-A-690323. These measurements shall meet the requirements of 3.8.3.

4.7.11 Antenna drive rotation test. Antenna Drive TG-176( )/GR and TG-177( )/GR shall be operated at 24 $\pm$ 0.5 VDC input voltage with Antenna Subassembly AS-2365( )/GR (H Adcock) and two extension shafts mounted. The rotation rates of the fixed pedestal and portable pedestal drive motor shafts shall meet the requirements of 3.6, 3.7.3 and 3.7.4 respectively when tested in accordance with paragraph 7.0 of SM-A-690323.

4.7.12 Output impedance measurement. The output impedance of Antenna Subassemblies AS-2360( )/GR through AS-2366( )/GR, inclusive, shall be measured with a suitable impedance bridge at each test frequency specified in Table VIII. The measurements shall meet the requirements of 3.5.5.5, 3.5.6.1 and 3.5.7.

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4.7.13 VSWR measurement (Log Periodic). The VSWR of Antenna Subassembly AS-2366( )/GR shall be measured over the frequency range of 190 through 1050 MHz in 50 MHz steps and shall meet the requirements of 3.5.7.

4.7.14 Frequency range test. Feed the output of a sweep generator into a whip antenna (or couple directly into the antenna under test). The sweep generator shall be capable of sweeping the frequency range of 0.535 to 1050 MHz. Antenna Subassemblies AS-2360( )/GR through AS-2366( )/GR inclusive, and their associated receivers, shall each be tuned over their entire frequency range. The sweep generator signal shall be picked up in the receiver audio output. The frequency range of the antenna subassemblies shall meet the requirements of 3.5.1.

4.7.15 Bearing accuracy test (Loops and H Adcock). The antenna test site shall be as specified in paragraph 4.2 of SM-A-690323. Determine the azimuth of the target antenna with respect to the test antenna using a transit or equal sighting device. Rotate the test antenna manually and determine the azimuth bearing, using the azimuth indicator display. This measurement shall be performed on Antenna Subassemblies AS-2360( )/GR through AS-2365( )/GR inclusive, and shall meet the requirements of 3.5.4 at the test frequencies specified in Table VIII. The loops shall be tested using vertically polarized target signals. The H Adcock shall be tested using vertically and horizontally polarized target signals. This test shall be performed at 24 $\pm$ 0.5 VDC input. This test shall also be performed at one frequency in each band using a 20 $\pm$ 0.1 VDC input.

4.7.16 Sensitivity measurement (Loops). The antenna test site shall be as specified in paragraph 4.2 of SM-A-690323. Orient the test loop for maximum response and tune for maximum sensitivity using the receiver listed in Table I. The target transmitter shall emit a 30 percent, 1000 Hz sine-wave modulated radio frequency signal, which is vertically polarized. The target output shall be adjusted for a difference in receiver audio output of 6 db for the condition of "MODULATION ON" to "MODULATION OFF". Further, the "MODULATION ON" condition shall correspond to 1 milliwatt. Once this condition is achieved, the field strength level existing at the receiving antenna shall be determined using the field strength meter (4.7.2.3). This level shall meet the requirements of 3.5.5.1. Sensitivity measurements shall be made at the test frequencies specified in Table VIII for Antenna Subassemblies AS-2360( )/GR through AS-2364( )/GR inclusive, in both sensed and unsensed modes. This test shall be performed at 24 $\pm$ 0.5 VDC input. The test shall also be performed at one frequency in each band using a 20 $\pm$ 0.1 VDC input.

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4.7.17 Gain measurement (H Adcock and Log Periodic). All gain measurements shall be performed at a test site specified in paragraph 4.2 of SM-A-690323. The receivers specified in Table VIII shall be used as detection devices. The output of the receiver shall be connected to a power output meter. A target generator, as specified in 4.7.2.2, shall be used. A dipole, cut to the measurement frequency and having an impedance of 50 ohms ± 1% shall be connected to the receiver. The target generator shall be tuned to the receiver frequency and adjusted so that indication on the power output meter results in not less than 10 db signal-plus-noise to noise ratio when the receiver dipole antenna is oriented for maximum response. No receiver Automatic Gain Control (AGC) shall be used and care shall be taken in receiver gain adjustment to prevent over loading and/or serious non-linearity. The receiver dipole shall than be replaced by the antenna to be tested, oriented for a maximum response. The target generator shall then be varied to produce the same output on the power output meter as the reference dipole. The ratio of the two signal generator readings shall be indicative of gain or loss. Gain measurements shall be made at the test frequencies specified in Table VIII for Antenna Subassembly AS-2365( )/GR and over the frequency range of 190 through 1050 MHz in 50 MHz steps for Antenna Subassembly AS-2366( )/GR.

4.7.17.1 Polarization. The gain of Antenna Subassembly AS-2365( )/GR (H Adcock) shall meet the requirement of 3.5.6.2, when measured at the full horizontal and the full vertical aspect. The gain of Antenna Subassembly AS-2366( )/GR (Log periodic) shall meet the requirements of 3.5.7.2 when receiving signals of full horizontal and full vertical polarization.

4.7.18 Dynamic range test. Dynamic range measurements shall be made with signal strengths of 0.10 volt/meter and 0.25 volt/meter present at the loop antenna under test. These tests shall be performed at the middle frequency of those specified in Table VIII for Antenna Subassemblies AS-2360( )/GR through AS-2364( )/GR. Each loop RF amplifier shall be operative and shall meet the requirements of 3.5.3, 3.5.4, 3.5.5.4 and 3.5.5.6.

4.7.19 Intermodulation distortion products test. Antenna Subassemblies AS-2360( )/GR through AS-2364( )/GR shall meet the requirements of 3.5.5.7 when tested per Method CS-03 of MIL-STD-462, Notice 3. The sample tuned frequency, F_o, shall be 30% modulated with a 1 kHz signal. Two radiating loops shall be used in lieu of the three-port network

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specified in Method CS-03. The signal input level for locating F_o shall be the minimum signal level to yield normal sensitivity reference of 10 db signal plus noise to noise ratio. Each antenna subassembly shall be tested at one frequency within operating range.

4.7.20 Envelope distortion test. Envelope distortion of all antenna subassemblies shall be measured in field strengths of 0.10 volt/meter and 0.25 volt/meter and at the test frequencies specified in Table VIII using a distortion analyzer. The input shall be 100 percent 1000 Hz amplitude modulated signals. Envelope distortion shall conform to the requirements of 3.5.5.8.

4.7.21 H Adcock tilt test. Actuate the antenna control unit polarization tilt switch (3.7.7) with Antenna Subassembly AS-2365( )/GR mounted on the antenna drive. The rotary actuator shall be capable of tilting the H Adcock antenna head to both the horizontal and vertical positions and shall meet the requirements of 3.5.6.4 when tested in accordance with paragraph 6.0 of SM-A-690323.

4.7.22 Band switch test. Switch the bands on Antenna Control C-6183( )/GR. Measure the elapsed time between initiating band change and actuation of the stepping-switch relays in the AS-2360( )/GR antenna subassembly. The time for actuation shall meet the requirement of 3.5.5.2.

4.8 Temperature test.

4.8.1 Low temperature. The equipments, less the canvas cases, shall be tested in accordance with Method 502.1, Procedure 1, of MIL-STD-810. The storage temperature shall be -70°F and the operating temperature shall be -40°F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Tables IV and V.

4.8.2 High temperature. The equipment, less the canvas cases, shall be tested in accordance with Method 501.1, Procedure II, of MIL-STD-810. The operating temperature shall be 150°F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Tables IV and V.

4.9 Humidity. The equipment, including the canvas cases, shall be subjected to the humidity test specified in Method 507.1, Procedure II, of MIL-STD-810, with test measurements during Step 4 and the last 5 hour

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period of the fifth 48 hour cycle in Step 6. Inability of the equipment to meet the requirements of 3.11.2 shall constitute failure of this test. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Tables IV and V. At the completion of this test, Group A electrical tests of Specification MIL-C-3885 shall be performed on all cable assemblies. All panel controls and connectors shall be operated to ascertain proper function. There shall be no mechanical deterioration, as evidenced by corrosion of metal parts or binding of rotating parts. All canvas items shall be inspected for deterioration (seams, zippers and moisture fungus proofing treatment).

4.10 Altitude. The equipment, less the canvas cases, shall be tested in accordance with Method 500.1 of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Tables IV and V.

4.11 Rain test. The equipment, less the canvas cases, shall be tested as follows:

a. Dry at $150^{\circ} \pm 5^{\circ}F$ for 48 hours.

b. Condition for 4 hours at $77^{\circ} \pm 5^{\circ}F$ and 40 to 60 percent relative humidity.

c. Take pretest data. The equipment shall meet full specification performance for the measurements specified in Table VII.

d. Test the equipment in accordance with Method 506.1, Procedure 1, of MIL-STD-810 with the exception that the wind source shall be 20 miles per hour and the rate of rainfall shall be 2 ± 0.5 inches per hour. Complete final measurements within 1 hour after the rain has been discontinued.

4.12 Dust. The equipment shall be tested in accordance with Method 510.1, Procedure 1, of MIL-STD-810. Step 3 shall be performed immediately after reaching stabilization in Step 2. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Tables IV and V.

4.13 Vibration, internal. Internal vibration of the equipment identified in 3.11.8 shall be measured as follows, to determine conformance to the requirements of 3.11.8:

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(a) Secure the equipment (shock mounts, if any, blocked or removed) directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 Hz. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one Hertz and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.14 Bounce tests.

4.14.1 Bounce (vibration) test; Class L Units. The equipment listed in Table II as Class L units shall be tested in accordance with Method 514.2, Procedure IX, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Tables IV and V. During Part 1, all connections (such as cable assemblies) shall be in place. When practicable, parkage tester accessories shall be selected from those listed in Drawing SC-A-46439. The adapter plate shall be free of cracks and broken welds, and all unused holes, except the 2 inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate. Additional barriers used to safeguard personnel shall not restrain the fore-and-aft motion of the adapter plate against the backstop.

4.14.2 Bounce (vibration) test; Class F units. The equipment listed in Table II as Class F units shall be tested in accordance with Method 514.2, Procedure XI, of MIL-STD-810. Prior to testing, the equipment shall meet the specification performance for the measurements specified in Tables IV and V. When practicable, package tested accessories shall be selected from those listed on Drawing SC-A-46439.

4.15 Shock test, bench-handling. The equipment listed in Table II shall be tested in accordance with Method 516.2, Procedure V, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Tables IV and V.

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4.16 Shock test; drop. The equipment shall be tested in accordance with Method 516.2, Procedure II, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Tables IV and V.

4.17 Leakage (immersion test). Battery Box CY-3589( )/GR, as prepared for field transportation, shall be tested in accordance with Method 512.1, Procedure I, of MIL-STD-810.

4.18 Orientation test. The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angle of 20 ± 3 degrees. During inclinations in each plane, the equipment shall meet specified performance for the measurements listed in Table VII.

4.19 Shrinkage test. The canvas carrying and stowage cases shall be inspected for conformance to 3.3.2.

4.20 EMI test. The equipment shall be tested in accordance with MIL-STD-462, Notice 3(EL).

4.21 Salt fog. The equipment shall be subjected to the salt fog test specified in Method 509.1, Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. Inability of the equipment to meet the requirements of 3.11.10 shall constitute failure of this test.

4.22 Fungus. The equipment shall be subjected to the fungus test specified in Method 508.1, Procedure I, of MIL-STD-810. There shall be abundant growth colonization (6.4) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.11.11 shall constitute failure of this test.

4.23 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.12. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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(a) External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mounting.

(d) Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units and location of the mating parts on the plug-in assemblies or major units.

(e) Size and form of special threads.

4.24 Visual and mechanical inspection. Equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.25 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

4.26 Rough handling test (preparation for delivery). When rough handling test in accordance with MIL-P-116 is required by the contract (see 6.2 (d)) the operational inspection specified in Table VII shall be conducted to determine freedom from operational malfunction caused by the rough handling.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified (see 6.2 (c)).

5.1.1 Level A.

5.1.1.1 Cleaning. Components of Antenna Group OE-4( )/GR shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Components of Antenna Group OE-4( )/GR shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

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5.1.1.4 Unit packing. Each field pack containing components of Antenna Group OE-4( )/GR shall be individually packed method III in accordance with MIL-P-116 as follows: Stow the components of the antenna group within their respective field packs CW-( )/GR. Secure all flaps, packets and snap rings. Cushion each field pack on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, designed to protect all projections and absorb the shock of impact in handling and transit. Place each cushioned item within a snug-fitting, fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, variety SW, grade W5c. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservative application and unit packing. Each Antenna Group OE-4( )/GR shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.

5.1.3 Commercial. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2(c)). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. Antenna Group OE-4( )/GR, unit packed as specified in 5.1, shall be individually packed within a snug-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4; or PPP-B-585, class 3, style 2 or 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, Finish A.

5.2.2 Level B. Antenna Group OE-4( )/GR, unit packed as specified in 5.1, shall be individually packed within a snug-fitting box conforming to PPP-B-601, domestic type; PPP-B-621, class 1, style 4; or PPP-B-585, class 3, style 2 or 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds

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100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Commercial. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. Antenna Group OE-4( )/GR is a component of Radio Receiving Sets AN/TRQ-23( ) and AN/TRQ-32( ) which are communication direction finding systems designed to receive, record and determine the directional bearing of transmitted signals in the 0.5 MHz to 405 MHz frequency range.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

(c) Level A, Level B or Commercial preservation and packing (see Section 5).

(d) When rough handling is required.

(e) First article inspection:

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(1) Four sample units of each item cited in section 1 are required so that lengthy environmental tests can be conducted concurrently on sample units.

(2) First article pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required.

(i) Quantity of tools and running spare parts required.

6.3 Color. Color chip nos. 24087 and 25102 per FED-STD-595 for the finishes required by 3.3.3 may be obtained upon request of Commanding Officer, U.S. Naval Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

6.4 Definitions.

6.4.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.4.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.4.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.4.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

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6.4.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.4.6 Exfoliation. Exfoliation is corrosion along the grain bounderies of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.5 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: C-6183W/GR. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Color of textiles. Standard samples of Olive Drab No. 7 may be obtained from the Supply Division, Clothing Textile Directorate, Defense Personnel Support Center, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

PREPARING ACTIVITY
Army (EL)

CUSTODIAN
Army(EL)

PROJECT NUMBER
5895-A-161

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-A-55468A(EL) ANTENNA GROUP OE-4()/GR

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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